

Midterm Lab Quiz 1: Service Fees and Generating Unique Hash Value

EVEN Units (PC) only

Develop a Solidity smart contract that calculates a total service fee and generates a unique hash value based on specific input parameters(sample data).

First name: Jose

Middle Name: Balsa

Last name: Rizal

Service code: [1 or 2]

****1** : Consultation = 20,000 plus 15% tax

****2**: Documents = 3,000 plus 8% service charge plus 15% tax

Service Fee:

Hash Generation requirements:

Message format = last character of the first name, first character of the middle name, last character of the Lastname, service code, first digit of the service fee

Service code =1 use abi.encode

Service code =2 use abi.encodePacked

Deploy the contract using first account of the ganache if the service code = 1

Deploy the contract using your MetaMask main account is the service code = 2.

Provide clear, high-resolution screenshots of the following:

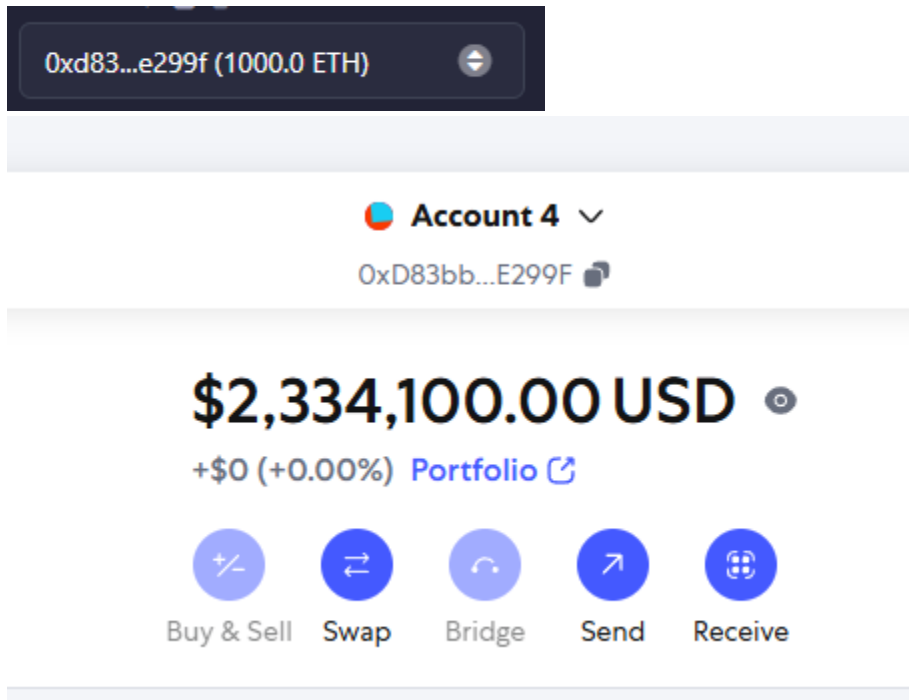
1. Wallet Balances: Your Ganache or MetaMask account before and after deployment to show the gas deduction.
2. Contract Execution: The Remix (or IDE) console showing the calculated Service Fee and the resulting Hash Value.

Code:

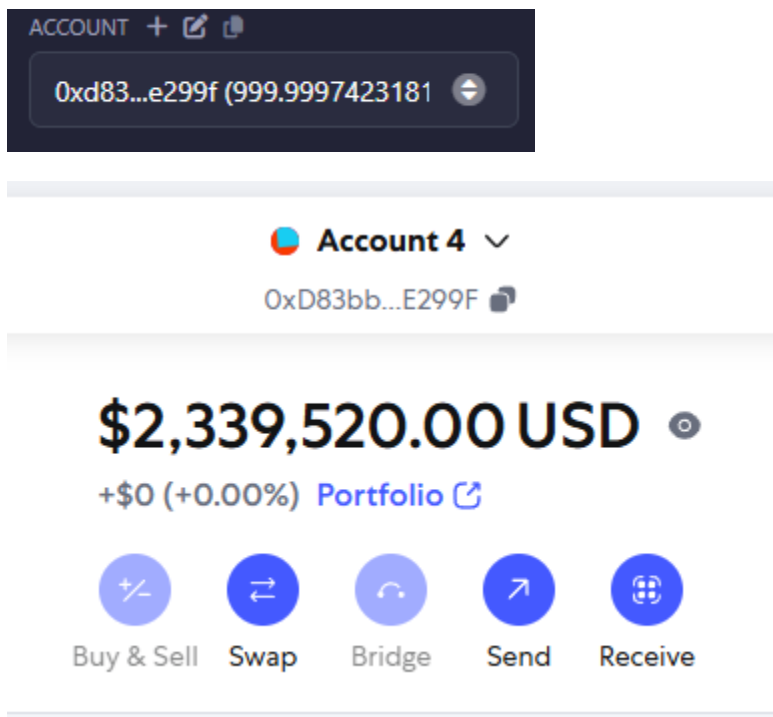
```
1  // Opiniano, Ezkylle
2  // NW - 301
3  // PC #38
4
5  // SPDX-License-Identifier: MIT
6  pragma solidity ^0.8.0;
7
8  contract ServiceFeeCalculator {
9      struct ServiceResult {
10         uint256 totalFee;
11         bytes32 hashValue;
12     }
13
14     uint256 constant CONSULT_FEE = 20000;
15     uint256 constant DOC_FEE = 3000;
16     uint256 constant TAX = 15;
17     uint256 constant SERVICE_CHARGE = 8;
18
19     function calculateService( infinite gas
20         string memory firstName,
21         string memory middleName,
22         string memory lastName,
23         uint8 serviceCode
24     ) public pure returns (ServiceResult memory) {
25
26         require(serviceCode == 1 || serviceCode == 2, "Invalid service code");
27
28         uint256 totalFee = serviceCode == 1
29             ? CONSULT_FEE + (CONSULT_FEE * TAX) / 100
30             : _documentFee();
31
32         bytes memory FirstName = bytes(firstName);
33         bytes memory MiddleName = bytes(middleName);
34         bytes memory LastName = bytes(lastName);
35
36         uint256 firstDigit = _firstDigit(totalFee);
37
38         bytes32 hashValue = keccak256(
39             abi.encodePacked(
40                 FirstName[FirstName.length - 1],
41                 MiddleName[0],
42                 LastName[LastName.length - 1],
43                 serviceCode,
44                 firstDigit
45             )
46         );
47
48         return ServiceResult(totalFee, hashValue);
49     }
50
51     function _documentFee() private pure returns (uint256) { infinite gas
52         uint256 withCharge = DOC_FEE + (DOC_FEE * SERVICE_CHARGE) / 100;
53         return withCharge + (withCharge * TAX) / 100;
54     }
55
56     function _firstDigit(uint256 value) private pure returns (uint256) { infinite gas
57         while (value >= 10) {
58             value /= 10;
59         }
60         return value;
61     }
62 }
```

Screenshot:

Before: Deploying the contract service code 1 on ganache wallet



After:



▼ SERVICEFEECALCULATOR AT 0

Balance: 0 ETH

CALCULATESERVICE

firstName: Ezkylle

middleName: P

lastName: Opiniano

serviceCode: 1

Calldata

Parameters

call

▼ SERVICEFEECALCULATOR AT 0

Balance: 0 ETH

CALCULATESERVICE

firstName: Ezkylle

middleName: P

lastName: Opiniano

serviceCode: 1

Calldata

Parameters

call

o: tuple(uint256,bytes32): 23000,0xaea46f3ca6455cddf87c30b53866496923dcbe7d260e2d9389d505524c524f2

Account 4 ▼

0xD83bb...E299F

\$2,339,519.40 USD

+\$0 (+0.00%) Portfolio



Buy & Sell



Swap



Bridge



Send

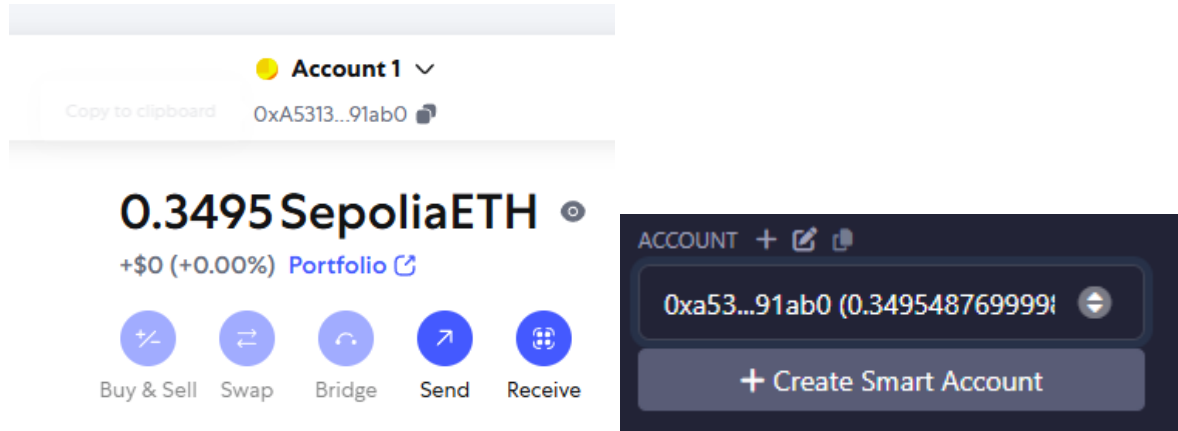


Receive

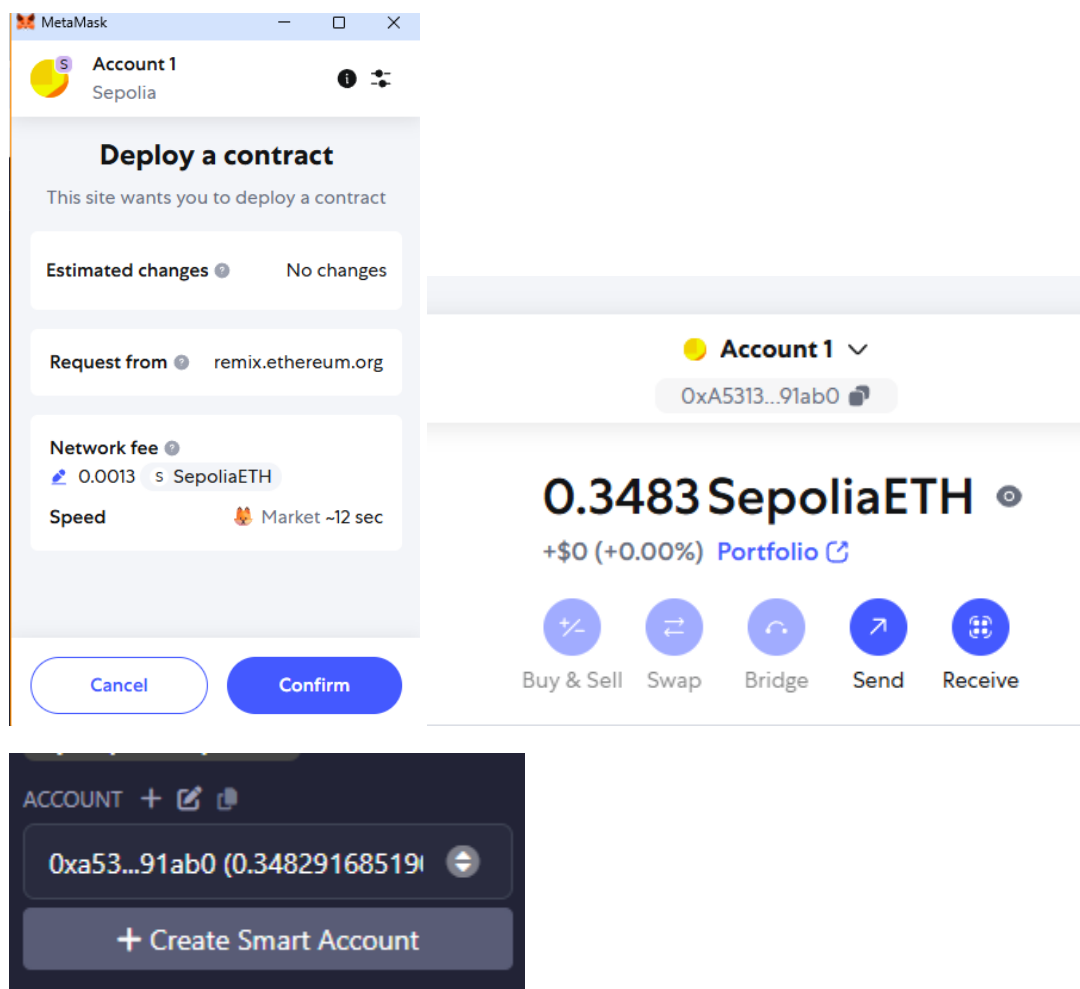
ACCOUNT +

0xd83...e299f (999.9997423181)

Before: Deploying the contract service code 2 on MetaMask wallet



After:



After input:

✓ SERVICEFEECALCULATOR AT 0

Balance: 0 ETH

CALCULATESERVICE

firstName: Ezkylle

middleName: P

lastName: Opiniano

serviceCode: 2

Calldata

Parameters

call

o: tuple(uint256,bytes32): 3726,0x0b0d1b4dd7c1398526b3da299090732608eaccddafa40c576c661fd83d0ea525

Account 1

0xA5313...91ab0

0.3483 SepoliaETH

+\$0 (+0.00%) Portfolio



Buy & Sell



Swap



Bridge



Send



Receive

ACCOUNT +

0xa53...91ab0 (0.34829168519)

+ Create Smart Account

